# Spatial Analysis

April 27, 2023

## 1 Methods

For each sample, we will compare the distribution of each markers in the normal and lesion regions.

- Markers for DD: CD3, CD4, CD8a, CD11c, CD15, CD45, CD68, CD69, CD123, CD279
- Markers for CF: CD1a, CD3, CD4, CD8a, CD11c, CD15, CD45, CD69, CD123, CD279 (CD68 did not pass QC)

For CD123, segmentation is done at flow=0.5 to get more masks in the bright area and a threshold of 5000 is set for total intensity to remove masks in the debris field in the lesion area.

The size of the skin tissue lesion area is determined by morphology using wheat germ agglutinin (WGA) flourescent stain. ImageJ is used to demarcate the lesion area, then a straight line ROI is drawn to span from one side of the lesion to the other side of the lesion, with the endpoints of the line positioned at the outermost skin boundary.

## 2 Results

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 74 | 1222 | 158 | 5395 | 631 | 2874 | 121 | 1399 |
| CD15 | 11 | 238 | 6567 | 2867 | 93 | 559 | 5050 | 743 |
| CD123 | 131 | 879 | 1350 | 8038 | 1117 | 2067 | 1038 | 2084 |
| CD11c | 343 | 2056 | 8029 | 17905 | 2926 | 4836 | 6174 | 4643 |
| CD68 | 154 | 998 | 2524 | 11081 | 1314 | 2347 | 1941 | 2874 |
| area | 0.12 | 0.43 | 1.30 | 3.86 | | | | |

Table 1: DD

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 422 | 8435 | 331 | 5629 | 976 | 4476 | 819 | 5574 |
| CD15 | 7 | 2154 | 756 | 391 | 16 | 1143 | 1871 | 387 |
| CD123 | 76 | 3680 | 324 | 1298 | 175 | 1952 | 801 | 1285 |
| CD11c | 624 | 8498 | 1308 | 4299 | 1443 | 4509 | 3237 | 4257 |
| area | 0.43 | 1.88 | 0.40 | 1.01 | | | | |

Table 2: CF

| | DD | CF |
|---|---|---|
| peaks distance - skin ref | 607 | 427 |
| peaks distance - DEJ ref | 319 | 337 |
| lesion width | 2862 | 1202 |

Table 3: Distance between CD3 peak and CD15 peak, measured in um.

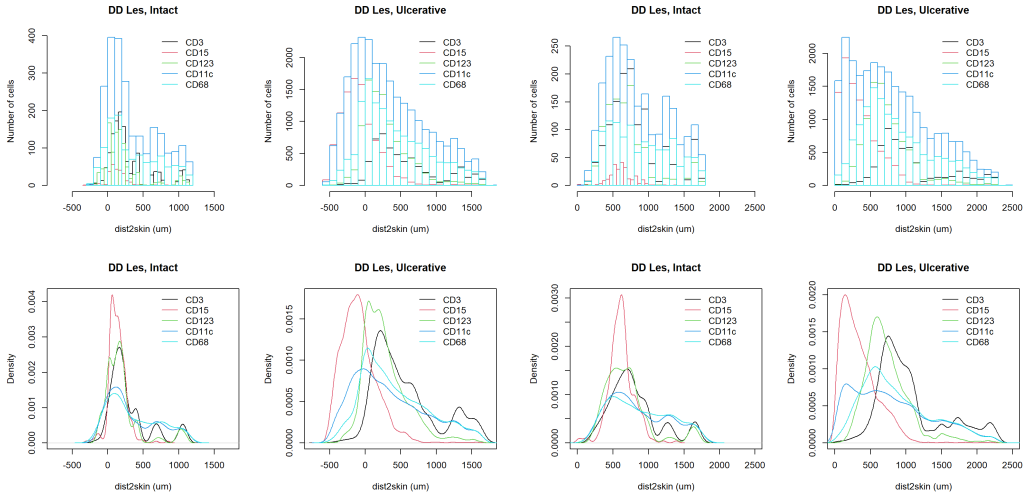



Figure 1: Distance to DEJ (left) and skin surface(right).

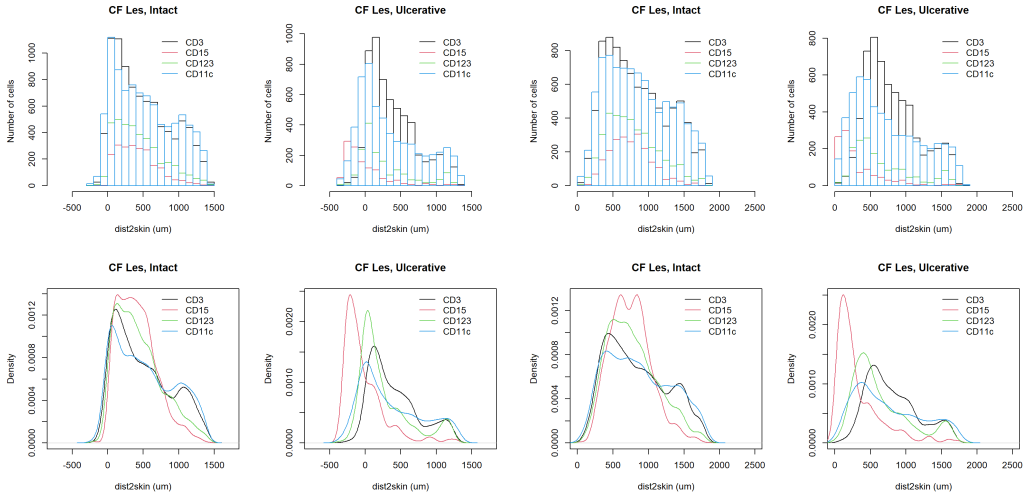



Figure 2: Distance to DEJ (left) and skin surface(right).